

a deep understanding of customers' needs and the ways to solve their business problems.

Digital thinking with digital literacy should be embedded within the new organisation. This can be done by becoming customer-focused and centric, and relying on data and feedback to drive the innovation.

In order to arrive at a bankable innovation, it is important to realise the digital value and reimagine what is possible to develop. It starts with strategy and then applies design thinking principles to develop a possible bankable innovation that comprises digital capability, productivity, platforms, and complex data management. The digital transformation needs of people, process, technology, and data analytics are further adopted and implemented.

Q. What are the challenges for startups in the post-pandemic era?

A. The businesses are having a tough time at the moment, but we need to keep getting inspired, keep inventing, and keep innovating. The major challenge for startups is to raise funds or get additional funds as there is a reluctance to invest from the funding organisations. For companies already in their digital journey mid-way, it is a big challenge.

We all know that the post-pandemic era will be more focused on the teams working from home, which may impact creativity, exchanging of ideas, energy levels, and the ability of teams to deliver and execute. Automation becomes another challenge.

So, we would recommend that startups need to work in a different way to address the above challenges. For instance, they can look into developing solutions of a smart city, which will cater to the post-pandemic requirement for social change.

This smart city solution can be powered by empathy for decision making, automation, artificial intelligence, machine-to-machine solutions, and emotional intelligence, apart from the defined needs of an IoT smart city solution. One positive observation is that breakthroughs in geographic barriers will happen as talent can be hired across the world for the project, but you need effective digital governance. **END** 

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What does he think of how the Indian FOSS ecosystem has shaped up? Ibrahim says, "Open source is one of the few things we have that is truly international. The contributors to any project cut across national, religious and racial lines. I think FOSS is one of the greatest achievements of human society, and I'm saying this without exaggeration. That's the global picture."

"On a local level, contributing to it requires things like a stable Internet connection, computing facilities, etc., We have made big strides in these areas in the past two or three decades in India. With programmes like Google Summer of Code and others, getting into open source is easier than ever, and a lot of Indian students are taking full advantage of this. If you go to a tier-1 or tier-2 city college, you can easily find lots of passionately driven folk who have contributed to open source," he adds.

Create a larger culture of tinkering and experimentation

"There are Indian organisations that are not limiting themselves to contributing to international open source projects. They are starting many of their own and maintaining these at a very high level of quality. One that comes to mind is the Swathanthra Malayalam Computing project in Kerala that has done enormous work; it has attracted several students and fresh engineers into the world of open source," he says.

But there is a big scope for improvement, as India needs to create a larger culture of tinkering and experimentation. Ibrahim says, "Back in my day, the first thing you'd be exposed to if you had a computer was a programming language, and 'using a computer' meant programming. These days, people are exposed to things that encourage them to be 'users'

rather than 'developers'. I think we should get young people comfortable with the idea that a computing device (a phone, laptop, or tablet) is a programmable device, and you can do several things with it beyond installing a simple app."

He adds, "Education should wire this in and should reward students even if they fail rather than focusing on achieving fixed results. Once that kind of curiosity is inculcated, all they need is a spark that can come from a random person or an online source to get them started on tinkering with software, and the problem will then be how to stop them. Unfortunately, our system focuses on 'following the rules' and that's a reasonably sure way of achieving mediocrity. If the next Elon Musk is going to be from India, he or she can be many things but being obedient is not a trait that comes to mind." **END** 